Thoughtful Labs AI Research Curriculum

Source Outline for the Living Residency Repository

Program Aim

Convert strong AI builders into research-capable practitioners through implementation, experiments, evaluation

Modules

1. Transformers From Scratch

Attention, tokenization, positional embeddings, decoding, and a runnable minimal transformer.

2. Training and Evaluation

Optimization, losses, reproducibility, baselines, ablations, and experiment reports.

3. Data, Embeddings, and RAG

Dataset design, embedding spaces, vector search, chunking, citations, and grounded generation.

4. Agents and Tool Use

Tool schemas, planning loops, memory, approval boundaries, trace review, and recovery.

5. Multimodal Systems

Vision-language and audio-language workflows, non-text evaluation, interfaces, and failure modes.

6. Evaluation and Observability

Benchmarks, judge models, golden sets, traces, regression gates, cost, and latency.

7. Alignment and Safety

Policy design, preference behavior, red teams, refusal quality, tool risk, and risk registers.

8. Research Methods

Literature review, replication, ablations, experiment logs, research memos, and peer review.

9. Systems and Deployment

Serving, APIs, monitoring, reliability, caching, cost modeling, and runbooks.